TASK 1

In [5]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [7]:
```
df=pd.read_csv("Dataset Internship.csv")
df
```

Out[7]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City
...	...	...	...	...	...	...
9546	5915730	Naml\ Gurme	208	××stanbul	Kemanke×× Karamustafa Pa××a Mahallesi, R\ht\m ...	Karak×_y
9547	5908749	Ceviz A××ac\	208	××stanbul	Ko××uyolu Mahallesi, Muhittin ××st×_nda×× Cadd...	Ko××uyolu
9548	5915807	Huqqa	208	××stanbul	Kuru×_e××me Mahallesi, Muallim Naci Caddesi, N...	Kuru×_e××me
9549	5916112	A×××k Kahve	208	××stanbul	Kuru×_e××me Mahallesi, Muallim Naci Caddesi, N...	Kuru×_e××me
9550	5927402	Walter's Coffee Roastery	208	××stanbul	Cafea××a Mahallesi, Bademalt\ Sokak, No 21/B, ...	Moda

9551 rows × 21 columns

In [8]:
```
print("Dataset Shape: ", df.shape)
```

```
Dataset Shape:  (9551, 21)
```

In [9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Restaurant ID         9551 non-null   int64  
 1   Restaurant Name       9551 non-null   object 
 2   Country Code          9551 non-null   int64  
 3   City                  9551 non-null   object 
 4   Address               9551 non-null   object 
 5   Locality              9551 non-null   object 
 6   Locality Verbose      9551 non-null   object 
 7   Longitude             9551 non-null   float64
 8   Latitude              9551 non-null   float64
 9   Cuisines              9542 non-null   object 
 10  Average Cost for two  9551 non-null   int64  
 11  Currency              9551 non-null   object 
 12  Has Table booking     9551 non-null   object 
 13  Has Online delivery   9551 non-null   object 
 14  Is delivering now     9551 non-null   object 
 15  Switch to order menu  9551 non-null   object 
 16  Price range           9551 non-null   int64  
 17  Aggregate rating      9551 non-null   float64
 18  Rating color          9551 non-null   object 
 19  Rating text           9551 non-null   object 
 20  Votes                 9551 non-null   int64  
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

In [10]:
```
missing_values=df.isnull().sum()
print("\n Missing values per columns",missing_values)
```

```
 Missing values per columns Restaurant ID           0
Restaurant Name         0
Country Code            0
City                    0
Address                 0
Locality                0
Locality Verbose        0
Longitude               0
Latitude                0
Cuisines                9
Average Cost for two    0
Currency                0
Has Table booking       0
Has Online delivery     0
Is delivering now       0
Switch to order menu    0
Price range             0
Aggregate rating        0
Rating color            0
Rating text             0
Votes                   0
dtype: int64
```

In [11]:
```
df.fillna({'Cuisines':'Unknown'},inplace=True)
```

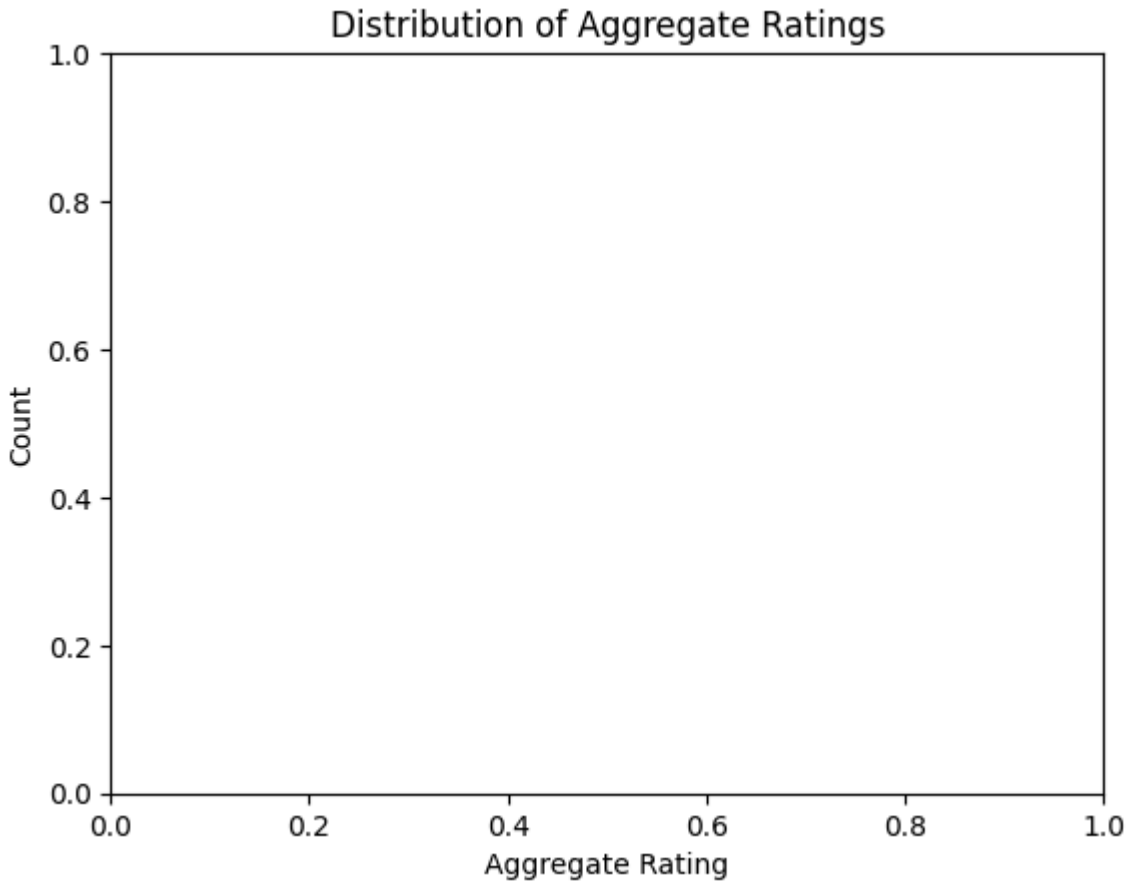
In [12]:
```
plt.figure(figsize=(8,5))
```

Out[12]:
```
<Figure size 800x500 with 0 Axes>
```

```
<Figure size 800x500 with 0 Axes>
```

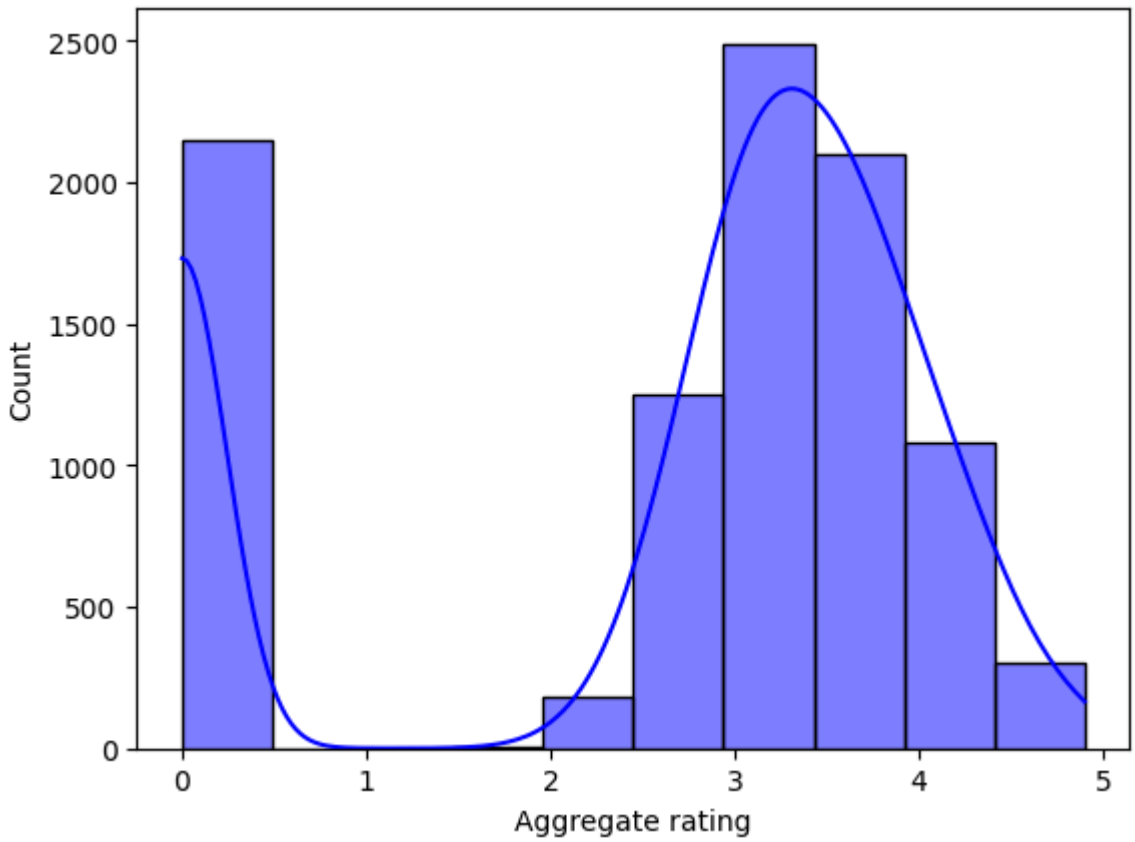
In [13]:
```
plt.title("Distribution of Aggregate Ratings")
plt.xlabel("Aggregate Rating")
plt.ylabel("Count")
```

Out[13]:
```
Text(0, 0.5, 'Count')
```



In [18]:
```
sns.histplot(df['Aggregate rating'], bins=10, kde=True,color='Blue')
```

Out[18]: <Axes: xlabel='Aggregate rating', ylabel='Count'>



In [20]:
```
ratings_counts=df['Aggregate rating'].value_counts().sort_index()
ratings_counts
```

Out[20]:
```
Aggregate rating
0.0    2148
1.8       1
1.9       2
2.0       7
2.1      15
2.2      27
2.3      47
2.4      87
2.5     110
2.6     191
2.7     250
2.8     315
2.9     381
3.0     468
3.1     519
3.2     522
3.3     483
3.4     498
3.5     480
3.6     458
3.7     427
3.8     400
3.9     335
4.0     266
4.1     274
4.2     221
4.3     174
4.4     144
4.5      95
4.6      78
4.7      42
4.8      25
4.9      61
Name: count, dtype: int64
```

In [22]:
```
df.to_csv("Final_Data.csv", index=False)
print("Process complete")
```

```
Process complete
```

In [ ]: